Investment and chronological operation of electricity storage and electrolysis

Methodology, Modelling and Consulting Skills
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Abstract

This study considers investment in batteries and electrolysers for the constructed energy system supplied in the project. It combines a deterministic investment appraisal (Tier 1) with a forecast-based chronological extension (Tier 2B). Before modelling, the five course datasets are reconciled to a common daily calendar. Missing generation records, query-endpoint snapshots, invalid demand observations and price-aggregation errors are repaired using archived upstream records, calibrated alternative series and documented imputation. Research on equipment costs and hydrogen support values provides a common financial basis.

Tier 1 uses mixed-integer optimisation to compare economic dispatch with supply-first operation over six historical years. Both policies choose electrolysis and zero batteries. A 10 t/day hydrogen outlet supports about 23 MW, whereas a national consumption ceiling supports hundreds to thousands of MW and illustrates the risk of using national demand to size a local asset. Replaying high-volatility periods and varying equipment costs and price dispersion tests when the investment would change.

The Tier 2B specification couples multi-step forecasts of generation, demand and electricity prices to rolling-horizon optimisation with storage-state feedback. It separates forecast evaluation from decision evaluation and fixes investment before the test period. This draft contains the completed Tier 1 analysis and the Tier 2B method and evaluation specification; forecast and control experiment results remain to be added.

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I. Introduction and data

1 Introduction

The project concerns a client's existing generation assets and the investment and operating decisions that can increase their value [1]. We study batteries and electrolysers in Tier 1 and select Tier 2B to examine chronological operation when future conditions must be predicted.

Our initial audit identified synchronised zero-generation records, missing dates, midnight snapshots used as daily means, abnormally low demand records and zero-volume price placeholders. The repair methods are full-day upstream reaggregation, short-gap interpolation, calibrated cross-source reconstruction and flagged whole-day imputation. These methods are distinguished in the data audit rather than treated as a single class of replacement.

Tier 1 combines public market and equipment evidence with mixed-integer optimisation. We compare free allocation of electricity with priority for current sales. Gain is the increase in annual equivalent profit over no investment under the same rule. For an illustrative buyer accepting up to 10 tonnes of hydrogen per day, both policies select 23.4 MW of electrolysis and no batteries. Annual gain is £12.68m under economic dispatch and £2.44m under supply-first dispatch. We test how offtake, equipment costs, and historical and stressed price variation affect those choices.

Tier 2B adds forecasts of generation, demand and prices to a rolling-horizon operating model. It repeatedly plans several days ahead, executes only the current day's decision and transfers the resulting battery stock to the next day. Figure I.1 shows the data, investment, prediction and state-feedback connections. Forecast accuracy and realised operating value will be evaluated separately; the experimental protocol is specified in Chapter III.

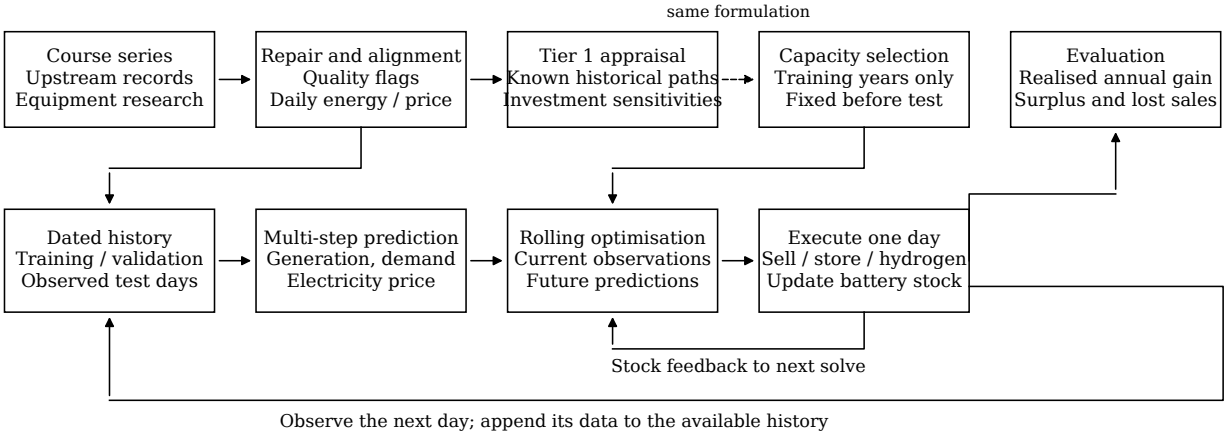


Figure I.1. Study architecture. Tier 1 uses known historical paths. Tier 2B uses past observations for forecasting and carries physical storage state forward; its investment decision precedes the test period.

2 Data audit and reconstruction

The five supplied files contain fuel-level generation, its total, simulated wind-farm generation, demand and market index prices. The fuel total is checked against its components; it is not added to them again. The separate simulated farm is added to that total as the declared asset boundary. Generation and demand are daily mean powers in MW, converted to MWh by multiplying by 24. We retain 1 January 2020–17 September 2026: 2,452 days in Coordinated Universal Time (UTC). Source checks use Elexon records and National Energy System Operator (NESO) series.

Table I.1. Detected problems, evidence and repair. Date counts can overlap.

Problem and extent	Identification and reconstruction
Eight non-wind/solar fuels zero on 107 days; solar zero on 42 days	Synchronous zeros conflict with archived records. Reaggregate Elexon actual generation by production type; use calibrated fuel half-hour or NESO solar series when that source is incomplete. Interpolate short gaps.
Midnight query endpoints: 7 generation, 350 demand, 346 price dates	Original values reproduce the first timestamp rather than the full day. Retrieve the complete UTC day and recalculate its mean.
Missing dates: 30 generation, 1 demand, 30 price	Calendar gaps are verified against the dated upstream series. Restore those records where available; otherwise use flagged whole-day estimates.
Invalid demand slots	Some five-minute records are below 10,000 MW, including 636 MW on 2 August 2022. Retain valid slots and reconstruct others from calibrated NESO demand; flag whole-day estimates.
Zero-volume price placeholders	115,571 provider records have no traded volume. Exclude those placeholders, volume-weight active providers within each half-hour, then average the day. Retain genuine zero and negative prices.

The repaired files share a complete calendar; fuel components sum to total generation, and the two price-unit columns retain their factor of 24. The 2 August demand mean is reconstructed as 28,744 MW. The audit identifies 436 dates with interpolation or estimation, 323 with cross-source reconstruction and 1,929 satisfying the strict quality flag; categories overlap [3]. The full set supports the primary analysis; the strict subset is a sensitivity check. All original values, replacements, source records and quality flags are retained. Historical prices are converted to 2020 pounds using monthly consumer price indices [4].

3 Common notation and assumptions

Only quantities shared by both tiers are defined here. Equipment decisions, dispatch policies, forecast information and technical parameters are introduced in their respective chapters.

Table I.2. Notation shared across the study.

Symbol	Meaning	Unit
$t \in \mathcal{T}$	UTC day in the selected study period	day
G_t, D_t	Available generation and market demand (sales ceiling)	MWh
p_t	Market electricity price, in constant 2020 pounds	£/MWh
S_t	Physical surplus, $\max(G_t - D_t, 0)$	MWh
Y	Represented years: sum of each sampled day's calendar-year fraction	year

The supplied series describe a constructed system rather than observed network congestion [1, 2]. Existing generation is fixed; its unchanged costs cancel in investment comparisons. Demand limits sales and is not an additional unpaid internal load. We assume no electricity imports, no additional export outlet and no payment for unused generation. Spatial transfer limits are omitted. All monetary comparisons use constant 2020 pounds. These conventions are common to both tiers.

II. Tier 1: deterministic investment appraisal

1 Problem analysis

We select battery and electrolyser capacities and their daily use over 2020–25 (2,192 consecutive days). The investment is made once and applies to all six years. Direct electricity sales provide the no-investment benchmark. The investment case depends on three questions: whether electrolysis can use electricity that could otherwise be sold, how much hydrogen a buyer can take, and whether storage can shift enough electricity to recover its costs.

2 Assumptions and research

2.1 Tier-specific assumptions

Policy A allocates electricity by economic value. *Policy B* serves current electricity demand first with generation and then with stored electricity. B honours this priority even at negative prices. Hydrogen is sold as produced subject to a common daily offtake limit. Battery electricity can feed electrolysis after any applicable sales priority. The daily model carries battery inventory forward, with at most one net charge or discharge per model per day and zero opening and closing usable inventory over the study period. Daily operation uses the historical paths.

We distinguish a broad market case and two smaller buyer cases. The broad cap of 1,559 t/day is the highest UK annual hydrogen consumption estimate in the 2022–24 source divided by the number of days in that year. The 1.56 and 10 t/day cases illustrate smaller outlets. Netback is the revenue per kg after non-electricity variable costs. We take £7.80/kg as a scenario, rebasing the UK policy benchmark of £175/MWh in 2012 pounds using 39.4 kWh/kg and the consumer price index. The national total includes existing uses and provides a scale comparison for the constructed system [8, 11, 4, 9].

2.2 Research objectives and parameter selection

Table II.1. Physical and financial inputs. Monetary values are in 2020 pounds.

Input	Model specification	Basis and use
Battery options	Four Sungrow, Huawei and Tesla models, 0.215–3.854 MWh and 0.108–1.927 MW per unit. Round-trip efficiency is 90–92%, state of charge is limited to 20–80%, and life is 15 years.	Supplier data for performance. Course and Danish Energy Agency costs for the smaller units and a public Tesla installed quote for the large unit. Capital is £55,129–1,239,440 per unit and annual fixed operation is £1,599–30,986 [1, 5, 12].
Electrolyser options	Eight ITM and Nel models, 1.31–50.31 MW. Specific consumption is 55.9–59.0 kWh/kg and the minimum online load is 10–25%, by model.	Manufacturer operating data. Common installed cost of £1.46m/MW lets us compare size and efficiency on the same financial basis [6, 12].
Electrolyser upkeep	£25,862/MW-year fixed operation, £489,609/MW per stack renewal in years 11 and 22, 30-year life.	UK cost study. A second case uses its complete cost basis with a reserve per kg of output in place of scheduled stack renewal [7, 12].
Financial treatment	5% real discount rate, zero residual value and battery commissioning charge.	Costs enter as annual equivalents. The 20% starting state of charge is accounted for through commissioning [9, 12].

3 Mathematical model

Indices i and j denote battery and electrolyser models. All daily energy flows and inventories are in MWh. Installed counts b_i, n_j and online electrolyser counts z_{jt} are nonnegative integers. The daily variables are direct sales q_t, generation sent directly to electrolysis g_t^H, battery charge c_{it}, discharge for sales r_{it} and for electrolysis u_{it}, usable stock s_{it}, and electrolyser input h_{jt}. The conversion factor κ_j is MWh/kg and L is the daily offtake limit in kg. With netback v in £/kg and annual equipment costs a_i^B, a_j^H, we maximise annual equivalent profit

$$\max \quad \Pi = \frac{1}{Y} \sum_{t \in \mathcal{T}} \left[p_t \left(q_t + \sum_i r_{it} \right) + \sum_j (v - w_j) \frac{h_{jt}}{\kappa_j} \right] - \sum_i a_i^B b_i - \sum_j a_j^H n_j. \tag{II.1}$$

Here w_j is the variable stack reserve in £/kg, zero when stack renewal is charged through the annual equipment cost.

The daily flow equations allocate generation, cap total sales at market demand and balance both sources of electricity into electrolysis:

$$q_t + g_t^H + \sum_i c_{it} \le G_t, \qquad q_t + \sum_i r_{it} \le D_t, \qquad \sum_j h_{jt} = g_t^H + \sum_i u_{it}. \tag{II.2}$$

The unallocated part of generation is unused. For battery i, W_i is the usable energy per unit, P_i the power in MW, and η_i the square root of round-trip efficiency. The inventory and daily throughput constraints are

$$s_{it} = (1 - \delta_i) s_{i,t-1} + \eta_i c_{it} - (r_{it} + u_{it})/\eta_i, \qquad 0 \le s_{it} \le W_i b_i, \tag{II.3}$$

$$c_{it} \le b_i \min(24 P_i, W_i/\eta_i), \qquad r_{it} + u_{it} \le b_i \min(24 P_i, \eta_i W_i). \tag{II.4}$$

Daily self-discharge δ_i is zero in the main case. We set $s_{i0} = s_{iT} = 0$ for $T = |\mathcal{T}|$, and at least one of c_{it} and $r_{it} + u_{it}$ is zero. A binary charge-mode indicator enforces this choice when needed. For electrolysis, H_j is the full-load daily input per unit and α_j its minimum operating fraction:

$$\alpha_j H_j z_{jt} \le h_{jt} \le H_j z_{jt}, \qquad z_{jt} \le n_j, \qquad \sum_j h_{jt}/\kappa_j \le L. \tag{II.5}$$

All flows are nonnegative. With $(x)_+ = \max(0, x)$, B also imposes

$$q_t = \min(G_t, D_t), \qquad \sum_i r_{it} = \min\left\{ (D_t - G_t)_+, \sum_i \eta_i (1 - \delta_i) s_{i,t-1} \right\}. \tag{II.6}$$

The short-duration battery models can discharge all usable stock within a day. A binary indicator selects the binding branch of the minimum.

For a unit with capital K, fixed annual operating cost F, lifetime N, real discount rate d and renewals C_k in years k, annual cost is

$$a = F + \frac{d}{1 - (1 + d)^{-N}} \left(K + \sum_k \frac{C_k}{(1 + d)^k} \right). \tag{II.7}$$

Reported gain subtracts each policy's own optimum with all $b_i, n_j = 0$.

Numerical checks. HiGHS certified the 14 portfolio cases with a maximum reported relative gap below 10^{-9}. Independent checks reproduced balances, cash flows and zero-battery dispatch. Six forced-storage cases tested stock carry-over, discharge to electrolysis and B's priority rule [12].

4 Results and analysis

4.1 Capacity and historical performance

In the broad market case, electrolysis-only and joint investment return the same portfolios. A selects 3,631 MW (721 Neptune V and two Neptune II units) and B selects 740 MW (147 Neptune V units). At 10 t/day, both choose 23.4 MW across six smaller modules. Initial capital in the broad case is £5.30bn (A) and £1.08bn (B), producing about 488,000 and 19,100 t/year of hydrogen respectively.

Table II.2. Six-year portfolio choices and annual profit gains by policy. A dash marks an excluded investment. Gains are in £m/year.

(a) Allowed assets at the broad market cap

	Policy A			Policy B		
Allowed investment	Battery (MWh)	Electrolysis (MW)	Gain	Battery (MWh)	Electrolysis (MW)	Gain
Battery only	0	–	0	0	–	0
Electrolysis only	–	3,631	1,639	–	740	37.71
Joint choice	0	3,631	1,639	0	740	37.71

(b) Smaller outlets with both assets allowed

Daily offtake	Battery	Electrolysis (MW)		Gain	
(t/day)	(MWh), A/B	A	B	A	B
1.56	0	3.84	3.84	1.91	0.35
10	0	23.42	23.42	12.68	2.44

At 10 t/day, B loses money in 2021, 2023 and 2024, with a worst year of £1.26m below no investment. These years have 43, 29 and 30 surplus days, respectively, against 154 in 2022. A can produce on additional days when the hydrogen value exceeds the price of forgone electricity sales. At the broad cap, A meets 82.2% of electricity demand and B meets 92.2%. Using the alternative UK cost basis at that cap leaves A's capacity unchanged and raises B's from 740 to 1,465 MW, with zero batteries in both cases [7, 12].

4.2 Energy attribution and contract sensitivity

The historical mean physical surplus is 2,276 GWh/year. For the zero-battery solutions we attribute $\min(g_t^H, S_t)$ to physical surplus first. The remaining electrolysis input uses energy within current electricity demand. We calculate the monetary loss separately against same-policy electricity sales without investment, accounting for negative prices under A. The aggregate records identify the available fuels by day and cannot trace the generating unit that supplied electrolysis.

Table II.3. Electrolysis sourcing and forgone electricity receipts. Energy is in GWh/year. The revenue reduction is in £m/year.

Offtake	Policy	Total input	Physical surplus	Within current demand	Electricity revenue reduction
Broad	A	27,255	2,184	25,070	1,618
Broad	B	1,067	1,067	0	0
10 t/day	A	183	43	140	9.18
10 t/day	B	43	43	0	0

In the broad market case, A uses about 2,184 GWh/year of physical surplus, only 8% of its total electrolysis input. Its electricity receipts fall by £1,618m/year. B uses 1,067 GWh/year of surplus and preserves direct sales, limiting its hydrogen throughput. At 10 t/day, each captures about 43 GWh/year, just under 2% of the physical surplus.

After reoptimising electrolysis and dispatch, the lowest netbacks that support any investment are about £3.32/kg in A and £4.57/kg in B at 10 t/day. At 1.56 t/day they rise to £3.38/kg and £4.66/kg. The 23.4 MW portfolio in Table II.2 is sized at £7.80/kg. Holding its operation fixed, it breaks even at £3.90/kg in A or £4.61/kg in B. At other prices, reoptimise operation and capacity.

4.3 Storage and investment sensitivity

Table II.4. Storage entry under historical and stressed conditions. Price multipliers scale the standard deviation of daily prices.

Test	Implication for storage investment
Historical high-volatility windows	All four battery options remain unselected in the highest-variance 60-, 90-, 180- and 365-day windows under both policies. The 90-day price standard deviation is £114/MWh against £65/MWh over six years.
Battery cost and price stress	Halving purchase and maintenance costs still gives zero batteries. With historical supply and demand, batteries first enter at about 40× (A) and 127× (B) the historical price standard deviation at the modelled costs.
Joint supply and price stress	The joint grid selects zero batteries at the historical price dispersion across nine supply–demand gap dispersions. At 120× price dispersion in B, raising gap dispersion from 1× to 1.1× changes the optimum from zero to 2.60 GWh.
Cavern compressed air	A separate 50 MW / 1.5 GWh test, at one-third of the reference battery installation cost per MWh and 62% round-trip efficiency, needs 5.38–5.50× (A) or 15.00–15.13× (B) price dispersion at the original supply–demand gap.

The volatility tests centre each series before scaling, preserving means and date order. The battery and cavern tests allow the electrolyser choice to adjust. For battery-only investment, even free charging and the full usable discharge on every positive-price day yield at most 57% of annual ownership cost across the four models. At historical prices the cavern's annual operating gain is £1.38m in A and £2.35m in B, against an annual cost of £8.52m [13].

4.4 Investment recommendations

Size electrolysis to secured demand. For 10 t/day, appraise the 23.4 MW portfolio against the £34.2m initial cost. A and B yield £12.68m and £2.44m/year respectively (Table II.2). Confirm the ability to displace electricity sales before relying on A. Under B, use the observed loss years to test cash reserves and the case for staged procurement. Reduce the appraisal to roughly 3.84 MW if the outlet is near 1.56 t/day. At a netback below the market entry levels above, defer electrolysis at the tested costs. The broad cap describes market scale. Capacity at that scale warrants appraisal only with much larger contracted offtake.

Defer daily storage. Both cost and historical stress tests support zero batteries (Table II.4). Reopen procurement if a lower installed cost or additional revenue brings the expected operating gain above annual ownership costs. For the cavern, the historical gain implies an annual cost ceiling of £1.38m under A or £2.35m under B on these inputs.

III. Tier 2B: forecasting and chronological operation

1 Information and experimental design

The operating decision is made once per day. Following the classroom simplification, current-day generation, demand and price are known at that decision, while future days are unknown [2]. Let $y_t = (G_t, D_t, p_t)$ and let $\mathcal{I}_t$ contain observations through day t, the installed equipment and the previous day's usable stock. Only information in $\mathcal{I}_t$ can determine the implemented action. Current-day aggregates are treated as observed at this daily planning decision.

We reserve 2020–22 for training and capacity selection, 2023 for model and horizon selection, and 2024–25 for the final test. A forecasting training target must lie before its corresponding cutoff. Repair calibration, scaling and hyperparameter selection use training/validation data only. The six-year Tier 1 capacity comparison remains a hindsight benchmark; test observations must not choose the capacity used in the predictive experiment.

2 Multi-step forecasting model

The proposed first predictive model is regularised linear autoregression, compared against a seven-day seasonal-naive predictor. At each forecast origin t, the feature vector f_t contains observed generation, demand and price lags, day-of-week indicators and annual sine/cosine terms. For variable $j \in \{G, D, p\}$ and horizon $k = 1, \ldots, H$, fit

$$\widehat{\beta}_{jk} = \arg\min_{\beta} \sum_{\tau \in \mathcal{A}_k} (y_{\tau+k,j} - f_\tau^\top \beta)^2 + \lambda \|\beta\|_2^2, \qquad \widehat{y}_{t+k,j} = f_t^\top \widehat{\beta}_{jk}, \tag{III.1}$$

where $\mathcal{A}_k$ contains eligible training origins whose targets are already observed, $\lambda \geq 0$ is the validation-selected penalty and H is the planning horizon. Forecast generation and demand are projected onto nonnegative values; negative price forecasts remain allowed. The seasonal-naive forecast uses the most recent observation with the same weekday, recursively if needed.

3 Rolling optimisation and storage feedback

At day t, installed battery and electrolyser counts are fixed. A planning problem uses observed values for the current day and predictions for $t+1, \ldots, t+H$. It retains the Tier 1 flow, demand, throughput and electrolysis constraints, but replaces the pooled energy balance with chronological stock:

$$s_\tau = s_{\tau-1} + \eta c_\tau - r_\tau/\eta, \qquad 0 \leq s_\tau \leq E_B b, \qquad \tau = t, \ldots, t+H. \tag{III.2}$$

Here s_{t-1} is the measured initial stock for that solve. The optimiser maximises forecast operating receipts plus a terminal-stock value; the latter is selected on validation data to avoid repeatedly liquidating storage at an artificial short horizon. Only (q_t, c_t, r_t, h_t) is implemented. Its realised stock becomes the next solve's initial condition. The test's initial and final stock treatment is held equal across all compared policies.

4 Evaluation and results to be completed

Forecast error is reported by series and horizon using mean absolute error and root mean square error, with price errors in £/MWh and physical errors in MWh. Operating evaluation compares the predictive controller, the same controller with seasonal-naive forecasts, a current-day rule and a full-path perfect-information optimiser, all at identical installed capacities and with identical data, costs and terminal stock. The gap to perfect information measures the realised loss from restricted information, rather than a change in investment scale.

Since zero batteries remove cross-day storage, fixed nonzero battery cases will diagnose prediction and feedback value alongside the training-selected investment. Results will include forecast errors, realised annual value, surplus capture, lost sales, stock feasibility and seasonal/adverse-period comparisons. These forecast and control experiments are pending.

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